

Communication

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Local observation of phase-segregation in mixed-halide perovskite

*Xiaofeng Tang,^{†1,4} Marius van den Berg,^{†2} Ening Gu,^{1,4} Anke Horneber,² Gebhard J. Matt,¹ Andres Osvet,¹ Alfred J. Meixner,² Dai Zhang*² and Christoph J. Brabec*^{1,3,4}*

¹Institute of Materials for Electronics and Energy Technology (I-MEET), Department of Materials Science and Engineering, Friedrich-Alexander-Universität Erlangen-Nürnberg, Martensstrasse 7, Erlangen, Germany

²Institute of the Physical and Theoretical Chemistry, University of Tübingen, Auf der Morgenstelle 15, Tübingen, Germany

³Bavarian Center for Applied Energy Research (ZAE Bayern), Haberstrasse 2a, Erlangen, Germany

⁴Erlangen Graduate School in Advanced Optical Technologies (SAOT), Paul-Gordan-Strasse 6, Erlangen, Germany

[†] These authors contributed equally to this work

* Correspondence and requests for materials should be addressed to:

C.J.B (christoph.brabec@fau.de);

D.Z (dai.zhang@uni-tuebingen.de);

ABSTRACT

Mixed-halide perovskites have emerged as promising materials for optoelectronics due to their tunable bandgap in the entire visible region. A challenge remains however in the photo-induced phase-segregation, narrowing the bandgap of mixed-halide perovskites under illumination thus restricting applications. Here we use a combination of spatially-resolved and bulk measurements to give an in-depth insight into this important yet unclear phenomenon. We demonstrate that photo-induced phase-segregation in mixed-halide perovskites selectively occurs at the grain boundaries rather than within the grain centers by using shear-force scanning probe microscopy in combination with confocal optical spectroscopy. Such difference is further evidenced by light-biased bulk Fourier-transform photocurrent spectroscopy, which shows the iodine-rich domain as a minority phase coexisting with the homogenously mixed phase during illumination. By mapping the surface potential of mixed-halide perovskites, we evidence the higher concentration of positive space charge near grain boundary possibly provides the initial driving force for phase-segregation, while entropic mixing dominates the reverse process. Our work offers detailed insight into the microscopic processes occurring at the boundary of crystalline perovskite grains and will support the development of better passivation strategies, ultimately allowing to process environmentally more stable perovskite films.

KEYWORDS: Perovskite, Photovoltaic, Photoluminescence, Phase-segregation, Scanning probe microscopy, Optical spectroscopy.

Introduction

Recent years have witnessed the promising rise of optoelectronic applications based on organometal halide perovskites (*i.e.* typically $\text{CH}_3\text{NH}_3\text{PbI}_3$) with the chemical formula ABX_3 , in which A is an organic cation, B is a metal cation and X represents the halide anion.^{1–4} A favorable feature of this semiconductor is attributed to its tunable bandgap, which can be straight forward done via tuning the halides' composition, allowing applications in color tunable light-emitting diodes (LEDs) and multi-junction or tandem solar cells.^{5,6} Partially substituting iodine with bromine is one of the most effective ways to realize bandgap tuning in perovskites, covering a range from 1.55 to 2.3 eV.⁷ Towards applications in tandem solar cells, especially as a top-cell above silicon or CIGS, mixed-halide perovskites can provide an optimum bandgap of 1.7–1.8 eV by tuning the ratio of I and Br into roughly 7:3 in the precursor.^{8,9} However, undesired phase-segregation was reported as one major obstacle when utilizing this material.⁹ When illuminated by resonant light, mixed-halide perovskites undergo phase-segregation, resulting in a lower band-gap absorption and red-shift photoluminescence (PL) which is characteristic for the iodine-rich phase.¹⁰ This strongly affects the voltage attained in mixed-halide perovskite-based solar cells and seriously restricts the applications.

Today, the majority of the efforts studying phase-segregation in mixed-halide perovskites relied on investigating steady-state bulk properties, like optical absorption and emission as well as X-ray diffraction (XRD).^{10–13} Only few spatially resolved investigations under restricted characterization conditions (*i.e.* stressed by vacuum and electron beam irradiation) were reported, suggesting nanoscale-sized iodine clusters as a likely product of a spatially restricted phase separation.¹⁴ However, the structure-property relations bridging from the microscopic to the macroscopic level remained unknown. It is widely accepted that ion migration goes hand-in-hand with phase-segregation, whereas ion migration is proposed to be strongly correlated with topography.^{15,16} Thus, localizing phase-segregation with high spatial resolution is of importance to gaining deeper understanding for this process.

In this work, we combine local-resolved and bulk investigations to identify phase-segregation in mixed-halide perovskites. Scanning probe microscopy combined with confocal optical spectroscopy offers the possibility to investigate the correlation between photo-induced phase-segregation and the local grain topography. Fourier-transform photocurrent spectroscopy (FTPS) confirms the formation of double bandgap features in the bulk of mixed-halide perovskites upon illumination. By

mapping the local surface potential, we propose the possible mechanism for the topography-dependent phase-segregation. Our findings stress the importance of the local grain topography on phase-segregation in mixed-halide perovskites and ultimately may support the development of novel concepts for overcoming the current restrictions of mixed-halide perovskite applications in photovoltaics.

Results and Discussion

So far, for investigating the local optical behavior of perovskite materials, reported investigations rely on scanning electron microscopy based cathodoluminescence imaging (SEM-CL)^{14,17} and confocal fluorescence imaging.^{18,19} However, the former method is performed under the stress of vacuum and electron beam irradiation, and the latter one does not provide comprehensive spectral information. As an alternative, we introduce scanning probe microscopy combined with confocal optical spectroscopy. It offers great advantages in revealing the local structural and compositional properties. By scanning a sample below a sharp metallic tip that is precisely positioned at the center of an optical focus, it is possible to directly correlate sample topography with optical emission at the same location. This way, the nanoscale structural and optical properties can be studied.^{20–23}

To verify the feasibility of this method, single-halide perovskites ($\text{CH}_3\text{NH}_3\text{PbI}_3$) samples were investigated first since this material is well studied and also more simple in its phase behavior as compared with mixed-halide perovskites. Figure S1a is a confocal optical image ($10\ \mu\text{m} \times 10\ \mu\text{m}$) of a $\text{CH}_3\text{NH}_3\text{PbI}_3$ film excited at 636 nm. The color code is reflecting the intensity of the photoluminescence signal. An inhomogeneous spatial distribution with a factor of about 1.5 is observed by the local variation of PL intensity. Additionally a hyperspectral PL image has been collected of the same region as Figure S1a, consisting of 32×32 spectra. Representative local spectra (Figure S1b) show well-resolved single emission peaks at 770 nm. Detailed analyses on the measured spots, such as full-width at half-maximum (FWHM) and PL peak position, are plotted against their position coordinates (Figure S1c and d). Respectively, 0.7 nm and 0.6 nm difference in FWHM and peak position show the local optical property of the $\text{CH}_3\text{NH}_3\text{PbI}_3$ film is overall uniform. The relatively minor inhomogeneity is deduced to mainly originate from the polycrystalline nature of $\text{CH}_3\text{NH}_3\text{PbI}_3$ thin film.^{18,19} We confirm that diffraction limited photoluminescence microscopy and spectroscopy is capable to resolve local optical properties of perovskite thin films.

We next investigate the phase-segregation of mixed-halide perovskites. In this study, a mixed-halide perovskite with the composition $\text{CH}_3\text{NH}_3\text{PbBr}_{3-x}\text{I}_x$ with $x=2.1$ was selected since its optical bandgap (*ca.* 1.77 eV) is ideal for applications in tandem devices onto Silicon.⁷ The $\text{CH}_3\text{NH}_3\text{PbBr}_{0.9}\text{I}_{2.1}$ perovskite is produced by spin-casting from precursor in one step (see Supporting Information for sample fabrication), presenting a heterogeneous distribution of crystal domains with median grain sizes of about 500 nm (see Figure S2 for the SEM image, bulk optical characterizations and XRD). Hoke et al. reported that $\text{CH}_3\text{NH}_3\text{PbBr}_{3-x}\text{I}_x$ perovskites have a rapid red-shift in PL and a splitting of XRD peaks due to photo-induced phase-segregation for bromide concentrations exceeding 20%.¹⁰ We confirmed the photo-induced phase-segregation in our samples by time-dependent PL as well as XRD equipped with external light illumination (Figure S3 and S4). Mixed-halide perovskites under light illumination showed the expected red-shift in PL and split of the XRD peaks. The PL evolved over time from 700 nm to 750 nm, and two distinguished emissions were observed during the transition. The phenomenon is reversible (Figure S4 and S5), which is consistent with previous reports.^{9,10} Additionally, since the typical XRD feature of PbI_2 is absence in Figure S4 and the PL spectrum (Figure S6) of PbI_2 shows negligible emission over the range of 700-750 nm, we exclude the possible formation of PbI_2 during illumination.

Locally, the spatial distribution of PL intensity for the mixed-halide perovskite $\text{CH}_3\text{NH}_3\text{PbBr}_{0.9}\text{I}_{2.1}$ film is shown in Figure 1a. Under the same measurement condition as for the pristine iodine compound ($\text{CH}_3\text{NH}_3\text{PbI}_3$), the PL intensity of mixed-halide perovskite now varies about by a factor of 4, which is 3 times larger than for $\text{CH}_3\text{NH}_3\text{PbI}_3$ perovskite (Figure S1a). To verify whether the origin for the inhomogeneous distribution of PL intensities is indeed halide phase separation, we further analyzed the local PL spectra. A significant distribution of bimodal PL features with the one emission peak at 1.77 eV and the other peak at 1.65 eV is observed (Figure 1b). For the $x=2.1$ perovskite, the emission at 1.77 eV comes from its bandgap. The red-shifted emission at 1.65 eV is attributed to the iodine-rich phases.¹⁰ In order to determine the spatial distribution of iodine-rich emission, we further analyzed a 32 x 32 hyperspectral PL image of Figure 1a by fitting each PL spectrum with two Gaussian peaks which are respectively located at 690-700 nm and 740-750 nm. These two peaks were assigned to the emission from “original” and “iodine-rich” bandgap regions, respectively.⁹ The spectrally integrated intensity of the two regions is shown in Figure 1c and d, respectively. The ratio of the two PL features ($\text{PL}_{740-750\text{nm}} / \text{PL}_{690-700\text{nm}}$) shows the proportion of PL from “iodine-rich”

domains at the measured spots (Figure 1e). The non-uniform distribution of PL intensity in “original” and “iodine-rich” domains indicates that phase-segregation in the mixed-halide perovskite film is spatially inhomogeneous. This seems to be in line with the widely accepted correlation that phase-segregation goes hand-in-hand with ion and defect migration.^{9,10} In polycrystalline perovskite films, ion migration is proposed to be non-uniform.¹⁵ We thus propose that photo-induced phase-segregation is also topography-dependent.

To investigate the dynamics of photo-induced phase-segregation as well as to verify our assumption, we tracked the local PL evolution by correlating shear-force scanning probe microscopy and diffraction limited PL spectroscopy. A direct correlation between topography and optical focus is ensured by accurately positioning the tip in the laser focus.^{24,25} A detailed experimental procedure can be found in the supporting information, Figure S7 and S8 as well as in our previous work.^{26,27} Figure 2a and b show the topography of a mixed-halide perovskite film. A heterogeneous distribution of crystal domains with median grain size of 450 nm is found, whereas the largest domains exceed 800 nm. As spatial-resolved PL spectroscopy focuses on a relatively small diameter of a few 100 nm, it is capable to resolve the spectral differences of the iodine-rich regions at the grain boundary. We afterwards monitor the evolution of the local PL emission spectra over time at selected sample positions (**i-vi**) for 600 seconds. Placing the focus of the laser spot on the center of grains (Figure 2, **i-iii**) shows a spectrally stable emission for 600 s. This is surprising, as bulk PL measurements (Figure S3a) do show photo-induced phase-segregation within such time periods.¹⁰

Contrarily, we find that the spectral PL evolution exclusively stems from the grain boundaries (Figure 2, **iv-vi**). The time evolutions of the intensity ratios $PL_{740-750\text{nm}}/PL_{690-700\text{nm}}$ (Figure S9) clearly show the variation in the proportion of PL from “iodine-rich” domains at the above measured spots. This result indicates that the phase-segregation undergoes to different extent with a strong impact from the local topography. The significant temporal evolution of the PL signal can be attributed to the regions at the edge of the grains, while PL from the grain center remains constant. We conclude that light illumination triggers phase-segregation dominantly at the grain boundaries, while the halide composition remains constant at the grain center with a stable band-edge PL emission. This significant difference highlights how closely photo-induced phase-segregation is related to the local topography. The red-shifted PL emission from the grain boundaries proves the existence and the photo-

induced formation of iodine-enriched domains. In direct contrast, the halide composition remains rather stable at the grain center with little changes in the alloy composition. We performed identical investigations on single-halide perovskites ($\text{CH}_3\text{NH}_3\text{PbI}_3$), and, as expected, observed neither at the grain boundary nor at the grain center any transient shifts in the PL (Figure S10). We record that confocal PL spectroscopy identifies grain boundaries as the dominant sites for photo-induced phase-segregation.

As photo-induced phase-segregation in mixed-halide perovskite is correlated to the microstructure, it appears sound to view illuminated mixed-halide perovskite as a blend rather than a uniform alloy. To further corroborate this point and provide a backup of the above local spectroscopic results, we performed bulk Fourier-transform photocurrent spectroscopy (FTPS) on a $\text{CH}_3\text{NH}_3\text{PbBr}_{0.9}\text{I}_{2.1}$ photoconductor device (device architecture in Figure S11a) with external white light illumination (Figure S11b). Originally developed to investigate sub-bandgap absorption in hydrogenated microcrystalline silicon, FTPS exhibits superior sensitivity.^{28,29} The major advantage of FTPS is its high sensitivity and wide spectral range (VIS-MIR). Unmodulated external light bias will not interfere with signal acquisition as the signal acquisition is in AC mode only.³⁰ This makes FTPS most suitable for the investigation of the band-edge of mixed-halide perovskite with respect to photo-induced phase-segregation. As shown in Figure 3a, the $\text{CH}_3\text{NH}_3\text{PbBr}_{0.9}\text{I}_{2.1}$ has a sharp onset at ≈ 1.77 eV and a spectrally well-resolved photocurrent. Within 10 min of white light soaking (Supporting Information), a new absorption shoulder at ≈ 1.65 eV is generated. The new shoulder vanishes after 1 h relaxation in the dark. The coming and going of the photocurrent shoulder indicates the reversible formation of a lower bandgap phase in mixed-halide perovskites under light soaking, which can be switched reversibly by illumination (Figure S12). In-situ FTPS (Figure 3b) shows the dynamics of the photo-induced bandgap evolution in $\text{CH}_3\text{NH}_3\text{PbBr}_{0.9}\text{I}_{2.1}$. The FTPS features show no detectable change after 10 min in prolonged light soaking experiments, which means that phase-segregation in illuminated mixed-halide perovskites reaches equilibrium. The reference measurements on single-halide perovskites ($\text{CH}_3\text{NH}_3\text{PbI}_3$ and $\text{CH}_3\text{NH}_3\text{PbBr}_3$) show negligible impact of light soaking (Figure S13). The increase in PL intensity is due to intrinsic defects (Figure S3b and c).^{19,31} We further deconvoluted the FTPS curve of illuminated mixed-halide perovskite, plotted in Figure 3c. It is clearly seen that the FTPS signal from the iodine-rich domain contributes about 10 % to the total FTPS intensity. Since the quantum efficiency of mixed-halide perovskites ($\text{CH}_3\text{NH}_3\text{PbI}_x\text{Br}_{3-x}$) are reported to vary not

significantly in the composition range between $2.1 \leq x \leq 3$,⁷ we assume that the total volume affected by halide phase-segregation is about 10 %, located along the grain boundaries. Certainly, phase-segregation will also produce a bromine-rich phases. Because of their relatively higher bandgap, we speculate that carrier may relax to lower energy sites before recombination. The existence of a second bandgap as seen from Figure 3a does not necessarily lead to a dramatically higher recombination rate. The iodine-rich phase with its smaller bandgap is located at the grains' boundary and thus may form a Type-I heterojunction to the larger bandgap of mixed iodine-bromine domain in the grains' center. Thus, by combining FTPS results with confocal PL spectroscopy, we can conclude that photo-induced phase-segregation selectively happens in mixed-halide perovskite at the grain boundaries (Figure 4a). The 1.77 eV bulk absorption bandgap is from the rather stable grain center (Figure 4b), and the 1.65 eV one is photo-generated by phase-segregation and iodine enrichment at the grain boundary (Figure 4c).

Since the FTPS investigation was conducted in a cryostat, it provides the possibility to investigate the influence of the ambient environment on the kinetics of phase-segregation. We tracked the FTPS evolution of mixed-halide perovskite during white light soaking under vacuum and in air (Figure S14). Within 30 min, we found negligible difference between the two spectra. Both the kinetics and the proportion of iodine-rich domains are almost identical. Thus, although environment was reported to strongly affect the long-term stability^{32,33} and photophysics^{34–36} of perovskites, we can conclude that the ambient environment barely impacts phase-segregation in mixed-halide perovskite. Additionally, we considered the possibility that electrostatic effects could produce the observed reversible structural changes and affect phase-segregation. Based on the theoretical investigation from Brivio et al., mixed-halide perovskites, $\text{CH}_3\text{NH}_3\text{Pb}(\text{Br}_x\text{I}_{1-x})_3$, are thermodynamically unstable against phase separation in the range of compositions between $x = 0.19$ and $x = 0.68$, the miscibility gap.³⁷ We therefore chose $\text{CH}_3\text{NH}_3\text{PbBr}_{0.3}\text{I}_{2.7}$ ($x = 0.1$), a theoretically stable alloy, to investigate its phase stability under light soaking. In Figure S15, the constant PL peak position and FWHM as well as the unvaried FTPS spectra both evidence that $\text{CH}_3\text{NH}_3\text{PbBr}_{0.3}\text{I}_{2.7}$ has a stable bandgap during light soaking, although electrostatic effect does exist in this perovskite material because of its ionic nature.^{38,39} Thus, the electrostatic effect can be deduced as a negligible factor for phase-segregation.

The possible mechanism of this topography-dependent photo-induced phase-

segregation can be explained by the excessive positive space charge close to the grain boundaries. As reported by Yun et al., grain boundary of polycrystalline (FAPbI₃)_{0.85}(MAPbBr₃)_{0.15} perovskite film has a lower surface potential than at the grain interior.⁴⁰ In our study, we found the same tendency in CH₃NH₃PbBr_{0.9}I_{2.1} (Figure 5a and b). The lower surface potential means that the energy levels are bending-up at the grain boundary (Figure 5c).⁴¹ This energy up-bending is indicative of a higher concentration of positive charges close to the grain boundary (Figure 5d).^{42,43} In the pristine state, negative charged ions are prone to diffuse to grain boundary for compensating the positive space charge, while the mixing entropy restrains this process. This trade-off guarantees uniformity of pristine mixed-halide perovskites in the dark. However, upon illumination, the excess photon energy provides sufficient energy for iodine ion migration. As recently reported by Draguta et al., the excited state of iodine-rich perovskite is calculated to have the lowest free energy (nearly zero) among all the possible compositions with single photon absorption.⁴⁴ This means that the formation of an iodine-rich domain in the photoexcited state is thermodynamically favoured. Consequently, iodine ions start to accumulate along a grain boundary and compensate the positive space charge by forming a meta-stable iodine-rich domain. We emphasize that this process exclusively happens near the grain boundary and not in the bulk or the surface. When turning off light, mixed-halide perovskite starts to relax back. The mixing entropy dominates as the driving force to redistribute iodine ions. Thus the mixed-halide perovskite is fully recovered to its pristine state.⁴⁴ The PL evolution of mixed-halide perovskite single crystals underpins the mechanism of grain boundary induced phase-segregation. Fresh mixed-halide perovskite single crystals with millimeter size were made using solution growth method.⁴⁵ PL spectra of single crystals were measured immediately after the single crystals being taken from the precursor. Under the equivalent condition as for thin film measurements, PL spectra of single crystals show a constant peak position and FWHM during illumination (Figure 5e), indicating that the mixed-halide phase is significantly more stable in single crystals compared to polycrystalline thin films. Noteworthy, the PL shoulder peak at longer wavelength range is reported as the optical reflection of the photoluminescence from the back surface of single crystal.⁴⁶ Since single crystal is virtually free of grain boundary, it provides strong backup of our above interpretation.

Overall, our findings confirm previous observations, which reported that mixed-halide perovskites with larger grain sizes are more stable during light soaking. We evidence that the reducing of the grain boundary density clearly suppresses photo-

induced phase-segregation in mixed-halide perovskite. Manipulation of grain boundaries was demonstrated to be an effective method to control charge carrier recombination, material stability as well as photovoltaic performance. Shao et al., reported open and fused grain boundaries showing obvious difference in affecting carrier recombination.⁴⁷ Park et al., showed the passivation of grain boundary by a molecular additive significantly improved the thermal stability of perovskite.⁴⁸ Therefore, exploring strategies how to treat grain boundaries to eliminate phase-segregation in mixed-halide perovskite will be the subject of future studies.

Conclusion

In conclusion, we report the mechanism for spatially and time-resolved photo-induced phase-segregation in mixed-halide perovskite. We find direct and unequivocal evidence that photo-induced phase-segregation in mixed-halide perovskites is selectively happening at the grain boundaries. No comparable phase separation was observed for the grain centers. Furthermore, FTPS investigations evidence two distinguished phases with two different bandgaps upon illumination, which iodine-rich phase acts as a minority phase coexisting with the pristine one. Therefore, illuminated mixed halide perovskites should be no longer viewed as a uniform alloy but rather as a blend, with light as the trigger to switch between the pristine and photo-generated states. The investigation of local surface potential provides the evidence of higher concentration of positive space charge at the grain boundary, which is proposed as the initial driving force for phase-segregation. The crucial role of grain boundary in phase-segregation was further confirmed by the stable performance of mixed-halide perovskite single crystals, which are virtually free of grain boundary.

The above observed results provide an in-depth understanding toward phase-segregation and an important guidance to direct further research and development for overcoming phase-segregation in mixed-halide perovskites. The correlation of topography with local confocal optics presented in this work offers an excellent platform for studying the spatial inhomogeneity of chemical and physical properties in polycrystalline photovoltaic materials.

Supporting Information

Materials and methods, Figures S1-S16.

Corresponding Authors

*E-mail: christoph.brabec@fau.de.

*E-mail: dai.zhang@uni-tuebingen.de.

Author Contributions

X.T and M.v.d.B. contributed equally.

Notes

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Figures

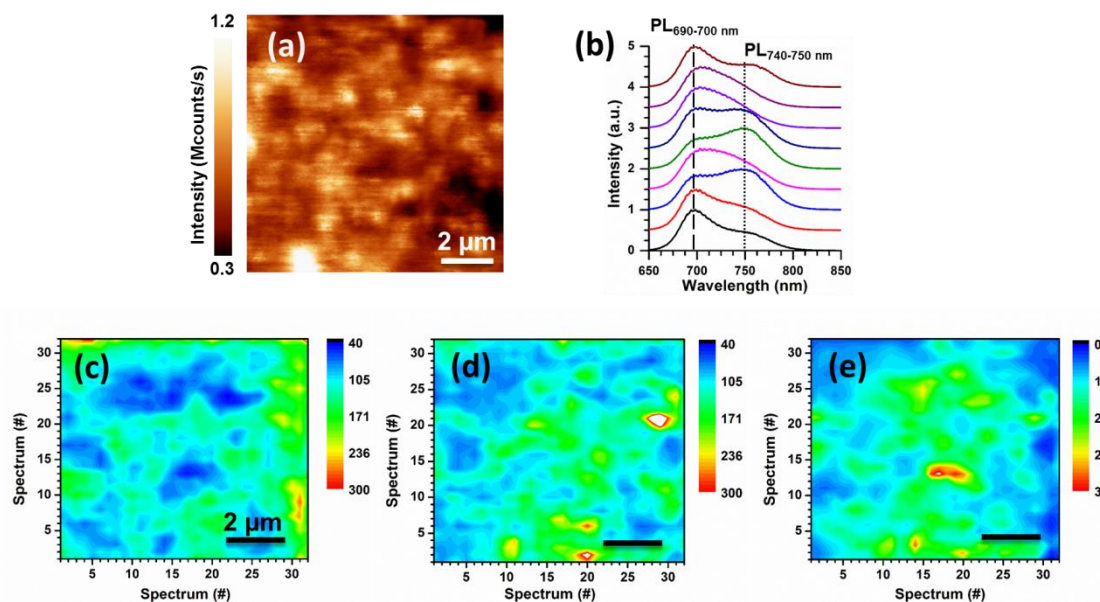


Figure 1. Local investigations on a CH₃NH₃PbBr_{0.9}I_{2.1} thin film. (a) Optical image, (b) representative local PL spectra, (c-e) 32 x 32 hyperspectral PL image of CH₃NH₃PbBr_{0.9}I_{2.1} perovskite with integrated region of the (c) 690-700 nm, (d) 740-750 nm and (e) ratio of PL_{740-750 nm}/PL_{690-700 nm}. Excitation wavelength at 636 nm with the intensity of 500 W/cm².

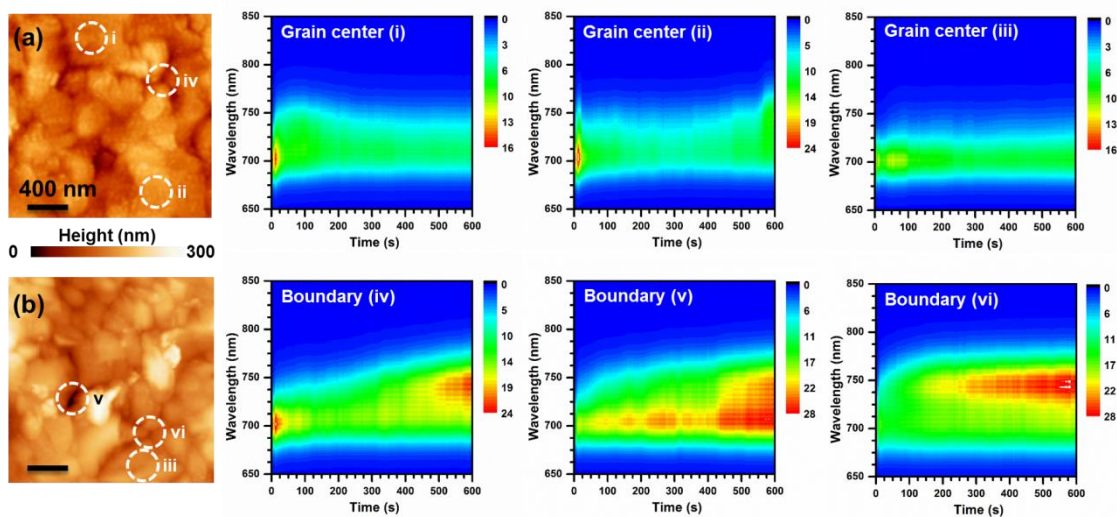


Figure 2. Local photoluminescence changes over time. (a and b) Topography of $\text{CH}_3\text{NH}_3\text{PbBr}_{0.9}\text{I}_{2.1}$ perovskite. PL evolution over time at different regions marked in a and b (i-iii represent grain center, iv-vi represent grain boundary). Excitation wavelength at 636 nm with the intensity of 500 W/cm^2 .

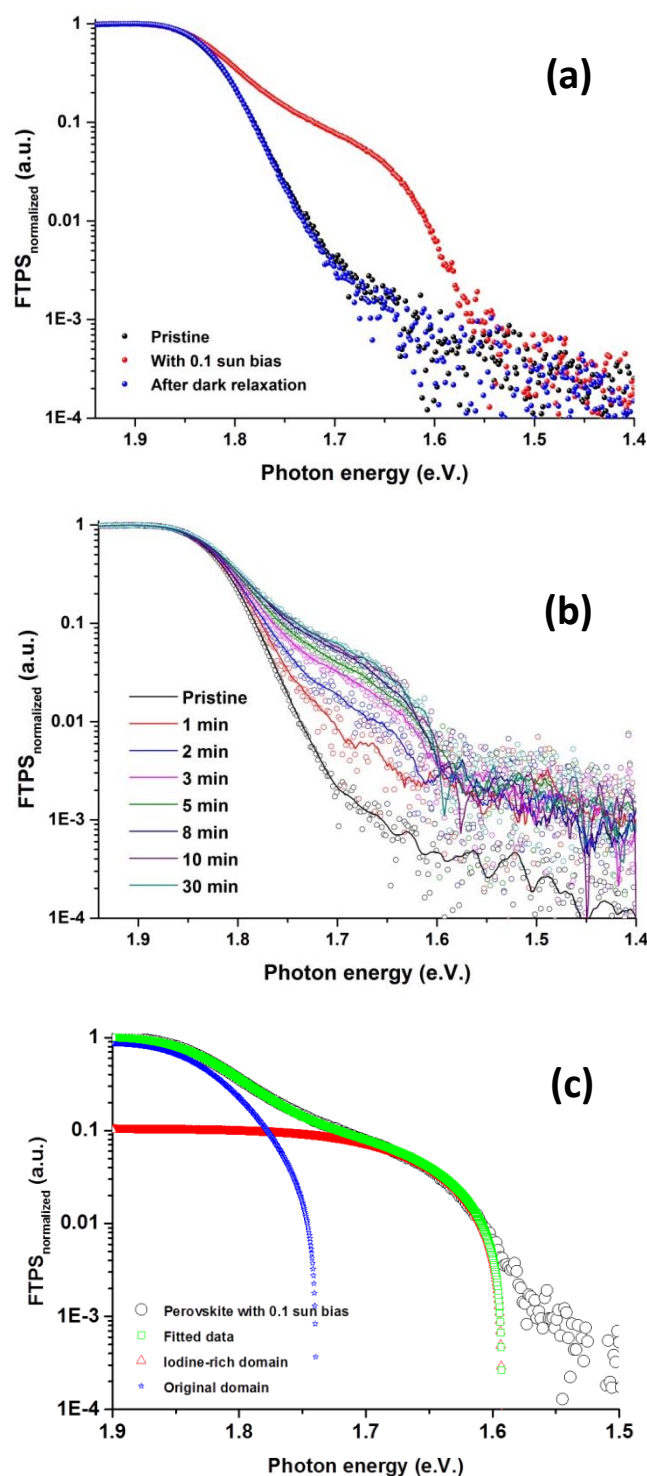


Figure 3. (a) FTPS of $\text{CH}_3\text{NH}_3\text{PbBr}_{0.9}\text{I}_{2.1}$ before (black) and after 10 min white-light soaking (red), and after 1 h relaxation in dark (blue). (b) In-situ FTPS of $\text{CH}_3\text{NH}_3\text{PbBr}_{0.9}\text{I}_{2.1}$ over time under white light soaking. (c) Fitting of the FTPS spectrum of $\text{CH}_3\text{NH}_3\text{PbBr}_{0.9}\text{I}_{2.1}$ after 10 min white-light soaking.

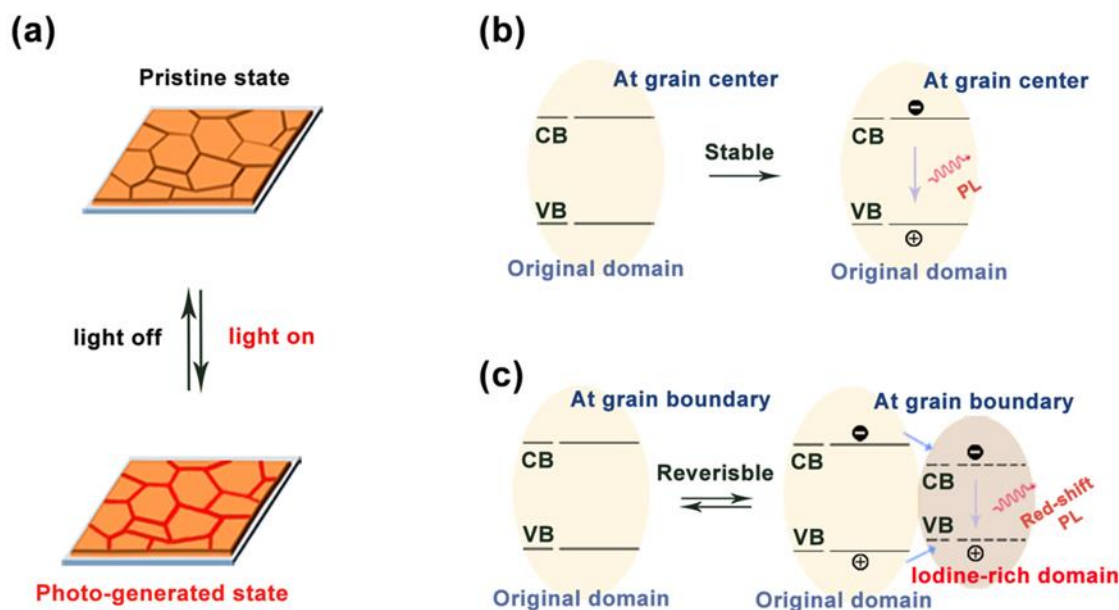


Figure 4. Schematics of the topography-dependent phase-segregation. (a) The external input of energy (light soaking) generates the reversible photo-induced phase-segregation in mixed-halide perovskite. Light herein acts as the trigger between the pristine state and the photo-generated state. The phase-segregation is highly topography-dependent. (b) At grain center, mixed-halide perovskite is rather stable during light soaking and reserves the original bandgap emission. (c) While phase-segregation is generated at grain boundary, forming the iodine-rich domain. The iodine-rich domain with the lowest bandgap acts as the charge recombination site and generates the red-shifted PL.

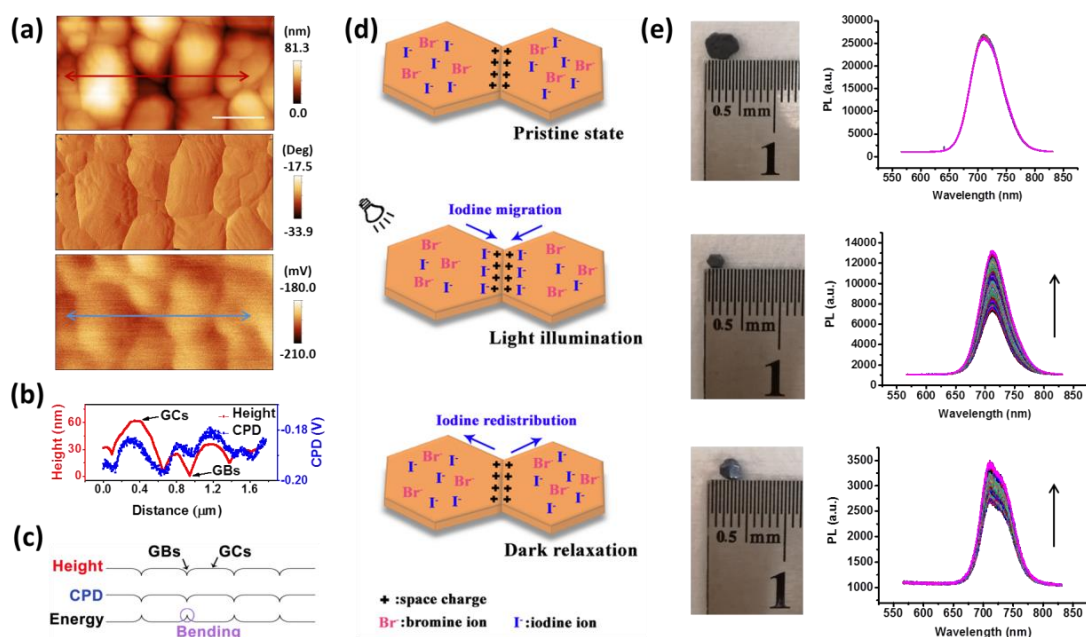


Figure 5. (a) AFM topography, phase and surface potential image of $\text{CH}_3\text{NH}_3\text{PbBr}_{0.9}\text{I}_{2.1}$. (b) Corresponding profile along selected lines labeled in topography and potential images. GCs represent grain centers, GBs are grain boundaries. (c) Energy alignment between grain boundary and grain center. (d) Schematic illustration of the possible mechanism for phase-segregation at grain boundary. Energy up-bending at grain boundary is indicative of a higher concentration of positive space charge near grain boundary. Under illumination, iodine ions start to migrate for compensating positive charges. After being relaxed in dark, accumulated iodine ions are prone to diffuse back and mix with bromine ions again in favor of the increasing of mixing entropy. (e) Photographs of millimetre-sized fresh $\text{CH}_3\text{NH}_3\text{PbBr}_{0.9}\text{I}_{2.1}$ single crystals and corresponding time-evolution of photoluminescence. Excitation wavelength at 450 nm with the intensity of 100 mW/cm^2 .

TOC

